

K-Nearest Neighbors — Predicting Workspace Amenity

Target & Predictors

Target: has_workspace — binary flag built from the *amenities* column (56% / 44% split — reasonably balanced)

Predictors: 10 numerical features spanning price, size, geography, demand, host scale

Sample: 1,946 rows after dropping missing values (588 dropped from 2,534)

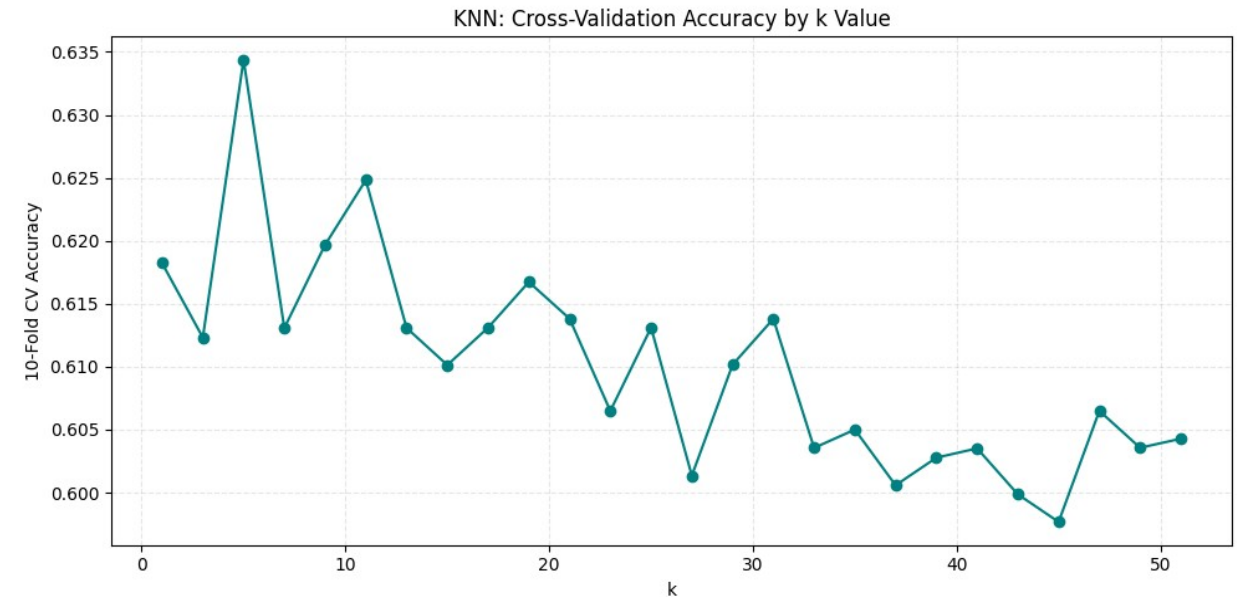
Method

- 70/30 stratified train-test split
- StandardScaler fit on training set only (distance-based model)
- 10-fold CV over odd $k \in \{1, 3, \dots, 51\}$ to avoid tied votes

k = 5

BEST k VALUE
10-fold CV accuracy: 63.4%

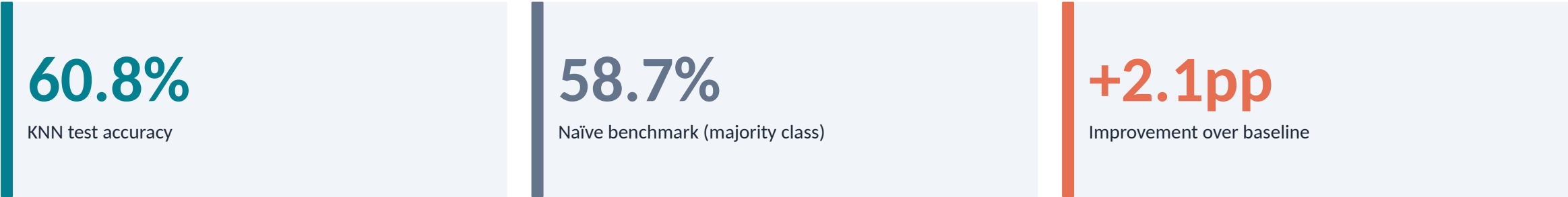
CV accuracy peaks sharply at $k = 5$, declines gradually thereafter



Predictor set (10 features)

price · accommodates · beds · latitude · longitude · number_of_reviews · availability_365 · minimum_nights · review_scores_rating · host_listings_count

KNN Results — Modest Lift Over Baseline



Confusion Matrix (test set, n = 584)

	Pred: No	Pred: Yes
Actual: No	88	153
Actual: Yes	76	267

Per-Class Performance

Class	Precision	Recall	F1	Support
No workspace	0.54	0.37	0.43	241
Has workspace	0.64	0.78	0.70	343
Weighted avg	0.59	0.61	0.59	584

TAKEAWAY

The model beats the majority-class baseline by only ~2 percentage points. Recall on listings WITH a workspace is strong (0.78), but recall on listings WITHOUT one is weak (0.37) — the model leans toward the majority class when the feature pattern is ambiguous. KNN provides some predictive signal, but limited.

Classification Tree — Predicting Host Response Time

Target & Predictors

Target: host_response_time — 4 classes, imbalanced (62% 'within an hour')

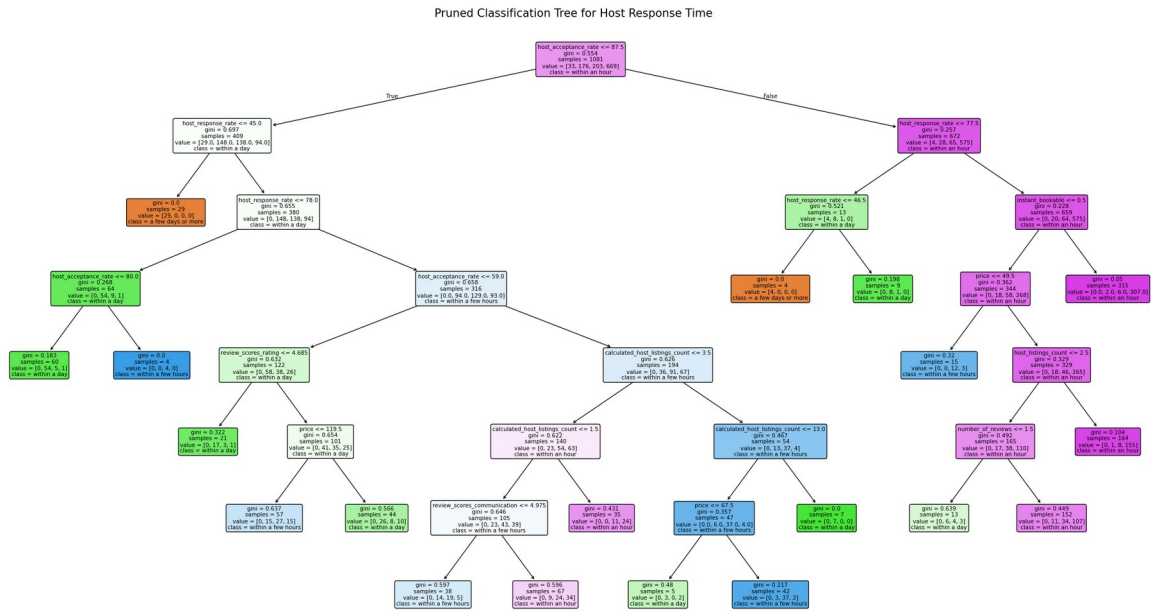
Predictors: 16 numerical features — host behavior, listing traits, demand, review scores

Sample: 1,545 rows; booleans mapped t/f → 1/0 for numerical splits

Tuning Strategy

- Grew full unrestricted tree on training data
- Extracted ccp_alpha candidates via cost_complexity_pruning_path()
- 10-fold CV over every alpha — picked the one maximising mean accuracy

Pruned tree — two host behavior variables drive the early splits



0.004

Best pruning α

7

Tree depth

19

Leaf nodes

75.8%

10-FOLD CV ACCURACY

at the optimal pruning level — strong enough to generalise, simple enough to avoid overfitting

Tree Results — 12-Point Lift and a Two-Variable Story

73.9%

Tree test accuracy

61.9%

Naïve benchmark

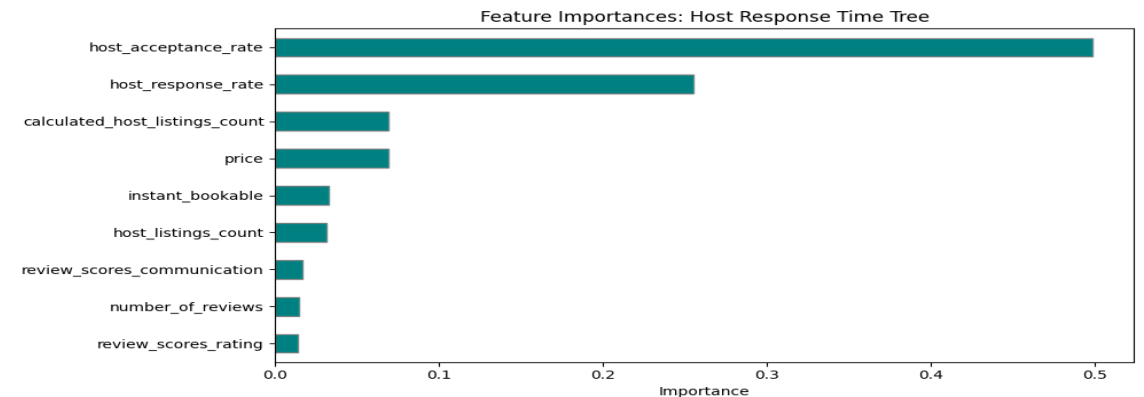
+12pp

Improvement over baseline

Per-Class Performance (test set, n = 464)

Class	Precision	Recall	F1	Support
a few days or more	1.00	1.00	1.00	14
within a day	0.59	0.56	0.58	75
within a few hours	0.46	0.26	0.33	88
within an hour	0.80	0.92	0.86	287

Feature Importances



TAKEAWAY

host_acceptance_rate (49.9%) and **host_response_rate (25.5%)** together drive 75%+ of all splits. The tree excels at the two extreme classes (F1 = 1.00 and 0.86) but struggles to separate the two middle categories — a common pattern with imbalanced ordinal targets.